1(a)	$h = 1.25$	B1
	$A \approx \frac{1}{2} \times 1.25 \left\{ 3.479 + 5.182 + 2\left(6.101 + 7.448 + 6.823\right) \right\}$	M1
	= 30.9	A1
		(3)
(b)	$\int_{0.5}^{5.5} \left(\mathrm{f}\left(x\right) + 4x\right) \mathrm{d}x = 30.9 + \left[2x^2 \right]_{0.5}^{5.5} = 30.9 + 2 \times 5.5^2 - 2 \times 0.5^2$ **or** $\int_{0.5}^{5.5} \left(\mathrm{f}\left(x\right) + 4x\right) \mathrm{d}x = \int_{0.5}^{5.5} \mathrm{f}\left(x\right) \mathrm{d}x + \frac{1}{2}\left(4 \times 0.5 + 4 \times 5.5\right) \times 5 = \ldots$	M1
	= 30.9 + 60 = 90.9	A1ft
		(2)
		Total 5

2(a)	$u_1 = 7 \Rightarrow u_2 = k - 7$	B1
	$u_2 = k - 7 \Rightarrow u_3 = k + k - 7 \Rightarrow u_4 = k - (2k - 7)$ $u_5 = k + (-k + 7) = 7$*	M1A1*
		(3)
(b)	$\sum_{r=1}^{4} u_r = 30 \Rightarrow 7 + k - 7 + 2k - 7 - k + 7 = 30 \Rightarrow k = ...$	M1
	$k = 15$	A1cso
		(2)
(c)	$\sum_{r=1}^{150} u_r = 37 \times 30 + 7 + "15" - 7$	M1
	$= 1125$	A1cso
		(2)
		Total 7

3(a)	$2(-3)^3-(-3)^2+A(-3)+B=55$ or e.g. $-54-9-3A+B=55$	M1
	$-54-9-3A+B=55$ $\Rightarrow 3A-B=-118$*	A1*
		(2)

(b)	$2\left(\frac{5}{2}\right)^3 - \left(\frac{5}{2}\right)^2 + A\left(\frac{5}{2}\right) + B = 0$	M1
	$3A - B = -118,\ 5A + 2B = -50$ $\Rightarrow A = ...,$ **or** $B = ...$	M1
	$A = -26,\ B = 40$	A1
		(3)

(c)	$\mathrm{f}(x)=(x-7)(2x^2+...x+...)+...$	M1
	$2x^2+13x+65$	A1
		(2)
		Total 7

4(a)	$y = 4x^{\frac{1}{2}} + 9x^{-\frac{1}{2}} + 3$ $\Rightarrow \left(\frac{\mathrm{d}y}{\mathrm{d}x} =\right) 2x^{-\frac{1}{2}} - \frac{9}{2}x^{-\frac{3}{2}}$	M1A1
		(2)
(b)	$\frac{\mathrm{d}y}{\mathrm{d}x} = 0 \Rightarrow 2x^{-\frac{1}{2}} - \frac{9}{2}x^{-\frac{3}{2}} = 0 \Rightarrow 4x - 9 = 0 \Rightarrow x = ...$	M1
	$x = \frac{9}{4}$ oe e.g. 2.25	A1
		(2)

4(c)(i)	$\left(\frac{d^2y}{dx^2} =\right) -x^{-\frac{3}{2}} + \frac{27}{4}x^{-\frac{5}{2}}$ oe e.g. $-x^{-\frac{3}{2}} + 6.75x^{-\frac{5}{2}}$	B1ft
(ii)	$\left(\left(\frac{d^2y}{dx^2}\right)_{x=\frac{9}{4}} =\right) -\left(\frac{9}{4}\right)^{-\frac{3}{2}} + \frac{27}{4}\left(\frac{9}{4}\right)^{-\frac{5}{2}} \left(=\frac{16}{27}(0.5925...)\right)$ $\left(\frac{d^2y}{dx^2}\right) > 0$ so (local) minimum	B1
		(2)
(d)	$0 < x < \frac{9}{4}$	B1ft
		(1)
		Total 7

5(a)	$(2+ax)^6 = 2^6 + \binom{6}{1}2^5(ax) + \binom{6}{2}2^4(ax)^2 + \ldots$	M1
	$= 64 + 192ax + 240a^2x^2 + \ldots$	A1A1
		(3)
(b)	$\left(3+\frac{1}{x}\right)^2 = 9 + \frac{6}{x} + \frac{1}{x^2}$ or $9 + \frac{3}{x} + \frac{3}{x} + \frac{1}{x^2}$	B1
	$f(x) = \left(9 + \frac{6}{x} + \frac{1}{x^2}\right)\left(64 + 192ax + 240a^2x^2 + \ldots\right) = \ldots$ Constant term is $9\times 64 + 6\times 192a + 240a^2$	M1
	$576 + 1152a + 240a^2 = 576 \Rightarrow 1152a + 240a^2 = 0$ $\Rightarrow 1152 + 240a = 0 \Rightarrow a = \ldots$	**d**M1
	$a = -\frac{24}{5}$	A1
		(4)
		Total 7

6	$2\log_4(x+1) = \log_4(x+1)^2$	B1
	e.g. $\log_4(12-2x) - \log_4(x+1)^2 = \log_4 \dfrac{12-2x}{(x+1)^2}$	M1
	$\dfrac{12-2x}{(x+1)^2} = 16$	A1
	$\Rightarrow 8x^2 + 17x + 2 = 0 \Rightarrow x = \ldots$	M1
	$x = -\dfrac{1}{8}$ oe e.g. –0.125	A1
		(5)
		Total 5

7(a)	$\left(u_{100}=\right)20+99\left(0.5\right)=\left(£\right)69.50*$	B1*
		(1)
(b)	$S_{300}=\frac{1}{2}\left(300\right)\left\{2\times 20+299\left(0.5\right)\right\}=...$ **or** $S_{300}=\frac{1}{2}\left(300\right)\left\{20+169.50\right\}=...$	M1
	$=\left(£\right)28425$	A1
		(2)
(c)	$20\times r^{299}=250\Rightarrow r=\sqrt[299]{\frac{250}{20}}\left(=1.008483032...\right)$	M1
	$S_{300}=\frac{20\left(1-r^{300}\right)}{1-r}=\left(27362.948...\right)$ $28425-27362.948...$	M1
	(£)1060	A1
		(3)
		Total 6

8(a)	$x^2+3=13-\dfrac{9}{x^2}\Rightarrow x^4+3x^2=13x^2-9$ $\Rightarrow x^4-10x^2+9=0$	M1A1
	$x^4-10x^2+9=0\Rightarrow\left(x^2-1\right)\left(x^2-9\right)=0\Rightarrow x^2=...$ $\Rightarrow x=...$	M1
	$x=1,\quad x=3$	A1
		(4)

(b)	$\int\left\{13-\frac{9}{x^2}-\left(x^2+3\right)\right\}\mathrm{d}x=13x+\frac{9}{x}-\frac{x^3}{3}-3x\left(+c\right)$ or $\int\left(13-\frac{9}{x^2}\right)\mathrm{d}x=13x+\frac{9}{x}\left(+c\right),\ \int\left(x^2+3\right)\mathrm{d}x=\frac{x^3}{3}+3x\left(+c\right)$	M1A1
	$\left[10x+\frac{9}{x}-\frac{x^3}{3}\right]_1^3=10\left(3\right)+\frac{9}{3}-\frac{3^3}{3}-\left(10+9-\frac{1}{3}\right)=...$ or $\left[13x+\frac{9}{x}\right]_1^3-\left[\frac{x^3}{3}+3x\right]_1^3=39+3-\left(13+9\right)-\left\{9+9-\left(\frac{1}{3}+3\right)\right\}=...$	**d**M1
	$=\frac{16}{3}$	A1
		(4)
		Total 8

9(a)	$\left(2\tan\theta = 3\cos\theta \Rightarrow\right) \dfrac{2\sin\theta}{\cos\theta} = 3\cos\theta$	M1
	$\dfrac{2\sin\theta}{\cos\theta} = 3\cos\theta \Rightarrow 2\sin\theta = 3\cos^2\theta = 3\left(1-\sin^2\theta\right)$	M1
	$2\sin\theta = 3\left(1-\sin^2\theta\right) \Rightarrow 3\sin^2\theta + 2\sin\theta - 3 = 0$*	A1*
		(3)
(b)	$\left(\sin\left(2x+\dfrac{\pi}{3}\right) =\right) \dfrac{-1\pm\sqrt{10}}{3}$ (May only see positive root) NB decimal roots are: –1.387…, 0.7207…	M1
	$2x+\dfrac{\pi}{3} = \sin^{-1}(0.7207...) \Rightarrow x = ...$	M1
	–0.121, – 2.50, 0.645, 3.02	A1A1
		(4)
		Total 7

10(a)	$x^2+y^2+4x-30y+209=0$ $\Rightarrow (x\pm 2)^2+(y\pm 15)^2 \ldots =0$	M1
(i)	Centre (–2, 15)	A1
(ii)	Radius $\sqrt{20}$	A1
		(3)
(b)	$y=mx+1 \Rightarrow (x+2)^2+(mx+1-15)^2=20$ or $y=mx+1 \Rightarrow x^2+(mx+1)^2+4x-30(mx+1)+209=0$	M1
	$x^2+m^2x^2+4x-28mx+180=0$ $b^2-4ac=0 \Rightarrow (4-28m)^2-4(1+m^2)\times 180=0$	**d**M1
	$(4-28m)^2-4(1+m^2)180=0 \Rightarrow 16-224m+784m^2-720-720m^2=0$ $\Rightarrow 2m^2-7m-22=0$*	A1*
		(3)
(c)	$2m^2-7m-22=0 \Rightarrow m=\frac{11}{2}, -2$	M1
	$m=\frac{11}{2} \Rightarrow \frac{125}{4}x^2-150x+180=0 \Rightarrow x=\frac{12}{5} \Rightarrow y=\frac{71}{5}$ or $m=-2 \Rightarrow 5x^2+60x+180=0 \Rightarrow x=-6 \Rightarrow y=13$	M1
	$\left(\frac{12}{5}, \frac{71}{5}\right)$ **or** $(-6, 13)$ oe	A1
	$\left(\frac{12}{5}, \frac{71}{5}\right)$ **and** $(-6, 13)$ oe	A1
		(4)
		Total 10

11(i)	E.g. $n = 5 \Rightarrow 3^5 + 2 = 245$	M1
	245 is not a prime number or e.g. 245 is divisible by 5 so not true	A1
		(2)

(ii)	$m = 3k+1 \Rightarrow m^2 - 1 = (3k+1)^2 - 1 = 9k^2 + 6k + 1 - 1$ **or** $m = 3k+2 \Rightarrow m^2 - 1 = (3k+2)^2 - 1 = 9k^2 + 12k + 4 - 1$	M1
	$m^2 - 1 = 9k^2 + 6k = 3(3k^2 + 2k)$ **or** $m^2 - 1 = 9k^2 + 12k + 3 = 3(3k^2 + 4k + 1)$	A1
	$m = 3k+1 \Rightarrow m^2 - 1 = (3k+1)^2 - 1 = 9k^2 + 6k + 1 - 1$ **and** $m = 3k+2 \Rightarrow m^2 - 1 = (3k+2)^2 - 1 = 9k^2 + 12k + 4 - 1$	**d**M1
	$3(3k^2 + 2k)$ and $3(3k^2 + 4k + 1)$ are both multiples of 3 so $m^2 - 1$ must be divisible by 3 when m is not divisible by 3	A1
		(4)
		Total 6